

# Does a Currency Hedge Neutralize Daily Yen Exposure?

## 货币对冲能否抵消日元的日度风险敞口？

A Multi-Factor Study of HEWJ versus EWJ Before and After the COVID-19 Break

疫情前后 HEWJ 与 EWJ 的多因子比较研究

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|               |                                                               |
|---------------|---------------------------------------------------------------|
| Course / 课程   | STA302 Final Project - Part 1 / STA302 期末项目 - 第一部分            |
| Prepared / 日期 | September 20, 2026 / 2026 年 9 月 20 日                          |
| Group / 小组    | Haiwen Yi; [add all other members / 补充其他成员]                   |
| Analysis / 分析 | R 4.4.3   2,753 daily observations   2014-02-06 to 2026-07-31 |

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### R rerun - headline evidence / R 重跑核心结果

|                         |         |                      |                 |
|-------------------------|---------|----------------------|-----------------|
| Full-model $R^2$ / 完整模型 | 71.21%  | Pre slope / 疫情前斜率    | -0.852          |
| Interaction / 交互项       | -0.0648 | OLS / HAC p          | 0.0043 / 0.0683 |
| Post slope / 疫情后斜率      | -0.917  | Maximum VIF / 最大 VIF | 3.55 (HML)      |

The yen-spread association is strong, but evidence of a post-pandemic change is sensitive to the standard-error method.  
日元与收益差的关系很强，但疫情后变化是否显著取决于标准误方法。

Before submission / 提交前：replace all bracketed member fields, align the contribution agreement, and add required OneDrive links. / 补全成员与贡献、统一小组协议，并添加课程要求的 OneDrive 链接。

## Part I - English proposal (official course version)

### Contribution statement

| Member                    | Proposed contribution                                                        |
|---------------------------|------------------------------------------------------------------------------|
| Haiwen Yi                 | Research design, data construction, R analysis, diagnostics, and first draft |
| [Member 2]                | [Specify literature, validation, writing, or presentation tasks]             |
| [Member 3]                | [Specify literature, validation, writing, or presentation tasks]             |
| [Member 4, if applicable] | [Specify contribution]                                                       |

The names and contributions must match the separately submitted group agreement.

### Introduction (341 words)

International equity investors face two return sources: the local equity market and the exchange rate that converts foreign assets into the investor's home currency. Currency-hedged exchange-traded funds attempt to remove the second component, but the hedge may be incomplete because contracts are rolled periodically, trading calendars differ, expenses and implementation costs exist, and the portfolio is not rebalanced continuously. This project asks: How strongly does daily yen appreciation explain the return difference between the currency-hedged iShares MSCI Japan ETF (HEWJ) and its unhedged counterpart (EWJ), and did that relationship change after the World Health Organization characterized COVID-19 as a pandemic?

The response is the daily log-return spread,  $(Y_t = 100[\Delta \log(\text{HEWJ}_t) - \Delta \log(\text{EWJ}_t)])$ . The focal predictor is  $(\text{JPYapp}_t = -100\Delta \log(\text{DEXJPUS}_t))$ , so a positive value denotes yen appreciation against the U.S. dollar. If the hedge fully removed contemporaneous currency exposure and both funds otherwise tracked the same Japanese equity portfolio, the expected slope would be near -1: yen appreciation should benefit the unhedged fund relative to the hedged fund. We estimate separate pre- and post-pandemic slopes through a categorical period indicator and its interaction with yen appreciation. March 11, 2020, the transition date, is omitted; the post period begins March 12.

The design is motivated by four peer-reviewed studies. Hau and Rey (2006) connect exchange rates, equity returns, and portfolio rebalancing, supporting joint attention to currency and equity movements. Campbell, Serfaty-de Medeiros, and Viceira (2010) show that optimal currency hedging varies with the investor and asset environment, so a mechanical one-for-one relation need not hold. Fama and French (2015) motivate controls for size, value, profitability, and investment factors. Shank and Vianna (2016) document dynamic links between exchange-rate movements and trading in U.S.-listed currency-hedged ETFs. Our contribution is narrower and transparent: a reproducible daily matched-pair analysis of one hedged/unhedged Japan ETF pair, with an explicit structural-break interaction, market-risk controls, and time-series-robust inference. The pandemic indicator is descriptive rather than a causal treatment because many policies and market conditions changed simultaneously.

### Data description (272 words)

The analysis uses 2,753 complete daily observations from February 6, 2014, through July 31, 2026: 1,348 pre-period and 1,405 post-period observations. Adjusted closing prices for HEWJ and EWJ come from the public Yahoo Finance chart history endpoints. The funds are a useful matched pair because iShares describes HEWJ as obtaining Japan equity exposure while hedging yen exposure and identifies EWJ as the underlying unhedged Japan equity fund. DEXJPUS, the noon New York yen-per-U.S.-dollar exchange rate, comes from the Federal Reserve Bank of St. Louis (FRED).

The response and three focal model terms are the return spread ( $Y_t$ ), yen appreciation ( $\text{JPYapp}_t$ ), the categorical Post indicator, and  $\text{JPYapp} \times \text{Post}$ . Eight additional predictors address observable market conditions: the daily Nikkei 225 return; Japan SMB, HML, RMW, and CMA factors; Japan momentum (MOM); the daily log change in VIX; and the previous month's U.S.-Japan short-term interest-rate differential. Japan factor returns are from Kenneth French's open Data Library and are

measured in U.S. dollars. The Nikkei 225, VIX, and interest-rate series are from FRED. Lagging the monthly rate differential by one month prevents using a same-month value that may not have been known at the start of each daily observation.

All series are joined by trading date, transformed only after alignment, and complete cases are retained. Percentage variables are expressed in percentage points, which makes the focal slope directly interpretable. The frozen raw files, SHA-256 checksums, cleaned CSV, R script, and machine-readable model outputs accompany this proposal. Data access links are: Yahoo HEWJ (<https://finance.yahoo.com/quote/HEWJ/history/>), Yahoo EWJ (<https://finance.yahoo.com/quote/EWJ/history/>), FRED DEXJPUS (<https://fred.stlouisfed.org/series/DEXJPUS>), FRED Nikkei 225 (<https://fred.stlouisfed.org/series/NIKKEI225>), FRED VIX (<https://fred.stlouisfed.org/series/VIXCLS>), and Kenneth French's Data Library ([https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data\\_library.html](https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html)).

## Ethics statement (194 words)

The project uses only public, aggregate market prices and factor series. It contains no personal information, human-subject data, private holdings, or individual trading records, so privacy and consent risks are minimal. Nevertheless, finance results can be misused if a descriptive association is presented as investment advice or as a causal pandemic effect. We will state that the results describe this ETF pair over this sample and do not establish that COVID-19 caused the slope change. Survivorship and product-selection concerns also matter: HEWJ and EWJ are currently traded funds selected because they form a convenient matched pair, not a random sample of all hedged products.

Reproducibility is addressed by preserving the exact downloaded files, documenting transformations, reporting the exclusion of March 11, 2020, and running the entire analysis in R. Adjusted prices incorporate provider-defined corporate-action adjustments; we will not imply that they are transaction prices available without costs. Yahoo, FRED, iShares, and Kenneth French data remain subject to their publishers' terms and revision practices. We will cite each provider and share links rather than claim ownership. Finally, any predictive evaluation is historical and does not support a promise of future returns or a trading recommendation.

## Preliminary results (368 words)

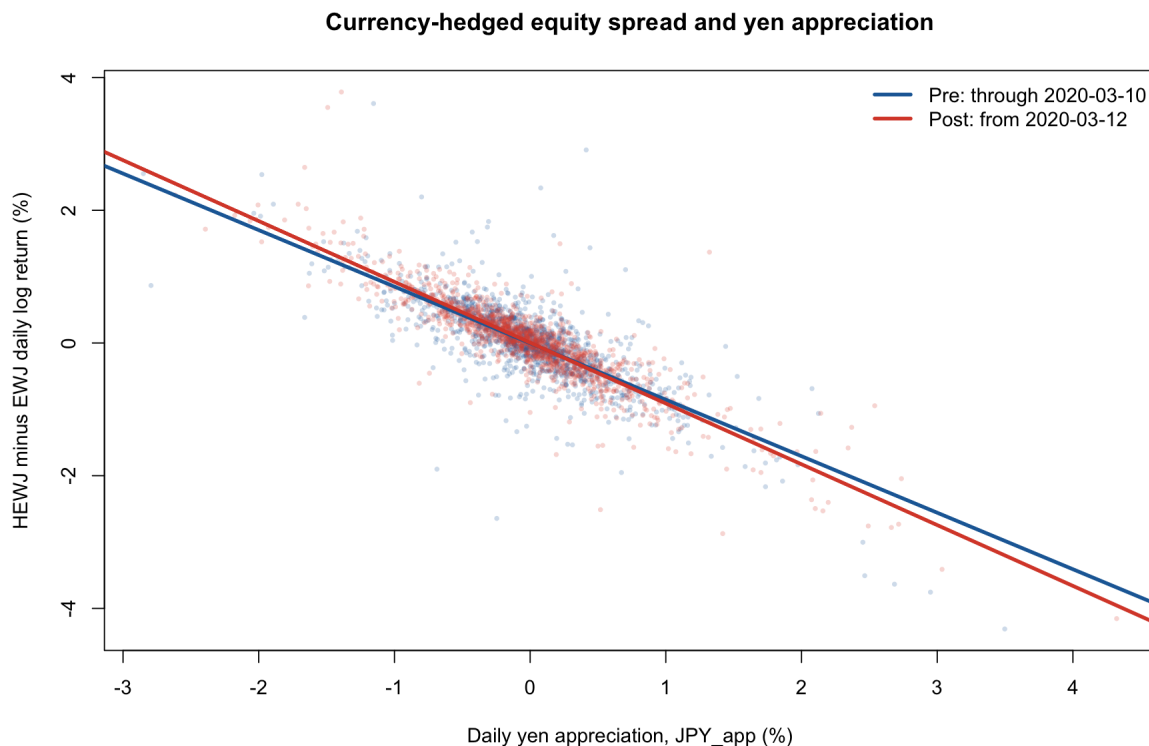


Figure 1. Daily HEWJ-minus-EWJ return spread versus yen appreciation. Lines are controlled period-specific slopes.

| Term                   | Estimate | Classical SE | Classical p | HAC(5) SE | HAC(5) p |
|------------------------|----------|--------------|-------------|-----------|----------|
| Yen appreciation (pre) | -0.8520  | 0.0178       | <0.001      | 0.0307    | <0.001   |
| Post indicator         | 0.0085   | 0.0147       | 0.562       | 0.0063    | 0.176    |
| Yen x Post             | -0.0648  | 0.0227       | 0.0043      | 0.0355    | 0.0683   |
| VIX log change         | -0.00687 | 0.00082      | <0.001      | 0.00164   | <0.001   |

Table 1. Selected full-model coefficients. HAC(5) changes inference for the pandemic interaction.

The full ordinary least-squares model is

$$Y_t = \beta_0 + \beta_1 JPYapp_t + \beta_2 Post_t + \beta_3 (JPYapp_t \times Post_t) + \gamma' Z_t + \varepsilon_t,$$

where ( $Z_t$ ) contains the eight market and factor controls. The model explains 71.21% of daily variation in the HEWJ-minus-EWJ spread (adjusted ( $R^2=71.10\%$ ); the focal-only model explains 70.37%. The pre-pandemic yen slope is -0.852 (classical SE 0.0178; HAC(5) SE 0.0307; HAC  $p < 0.001$ ). Thus, a one-percentage-point yen appreciation is associated with an approximately 0.852-percentage-point lower hedged-minus-unhedged return before the break, conditional on the controls.

The interaction estimate is -0.0648. Under classical OLS it is statistically significant (SE 0.0227,  $p = 0.0043$ ), but with Newey-West HAC(5) inference it is only marginal (SE 0.0355,  $p = 0.0683$ ; 95% CI [-0.1345, 0.0049]). The implied post-pandemic slope is -0.917. Both the pre and post slopes differ statistically from the theoretical -1 benchmark under HAC(5), although the post-period difference is economically smaller. The level shift at zero yen movement is 0.0085 percentage points and is not significant with HAC inference ( $p = 0.176$ ). Among the controls, the VIX change has a negative coefficient (-0.00687; HAC  $p < 0.001$ ); most other control coefficients are imprecisely estimated.

The model does not satisfy all ideal iid-error assumptions. The Breusch-Pagan test rejects homoskedasticity ( $p = 0.0048$ ), the five-lag Breusch-Godfrey test rejects no serial correlation ( $p < 0.001$ ), and the Q-Q plot shows heavy tails. Durbin-Watson is 2.85, consistent with negative first-order residual autocorrelation. Multicollinearity is not severe: the largest VIF is 3.55 (HML). These findings are why the main interpretation reports both classical OLS and HAC(5) uncertainty rather than relying on conventional p-values alone.

As a secondary predictive check, an 80/20 chronological split trained through January 29, 2024, and tested from January 30, 2024, through July 31, 2026. The full model achieved RMSE 0.313 and MAE 0.219 percentage points, compared with 0.654 and 0.463 for a training-sample historical-mean benchmark. This check is supportive but not a substitute for inference because later observations may come from a different regime.

## Residual diagnostics

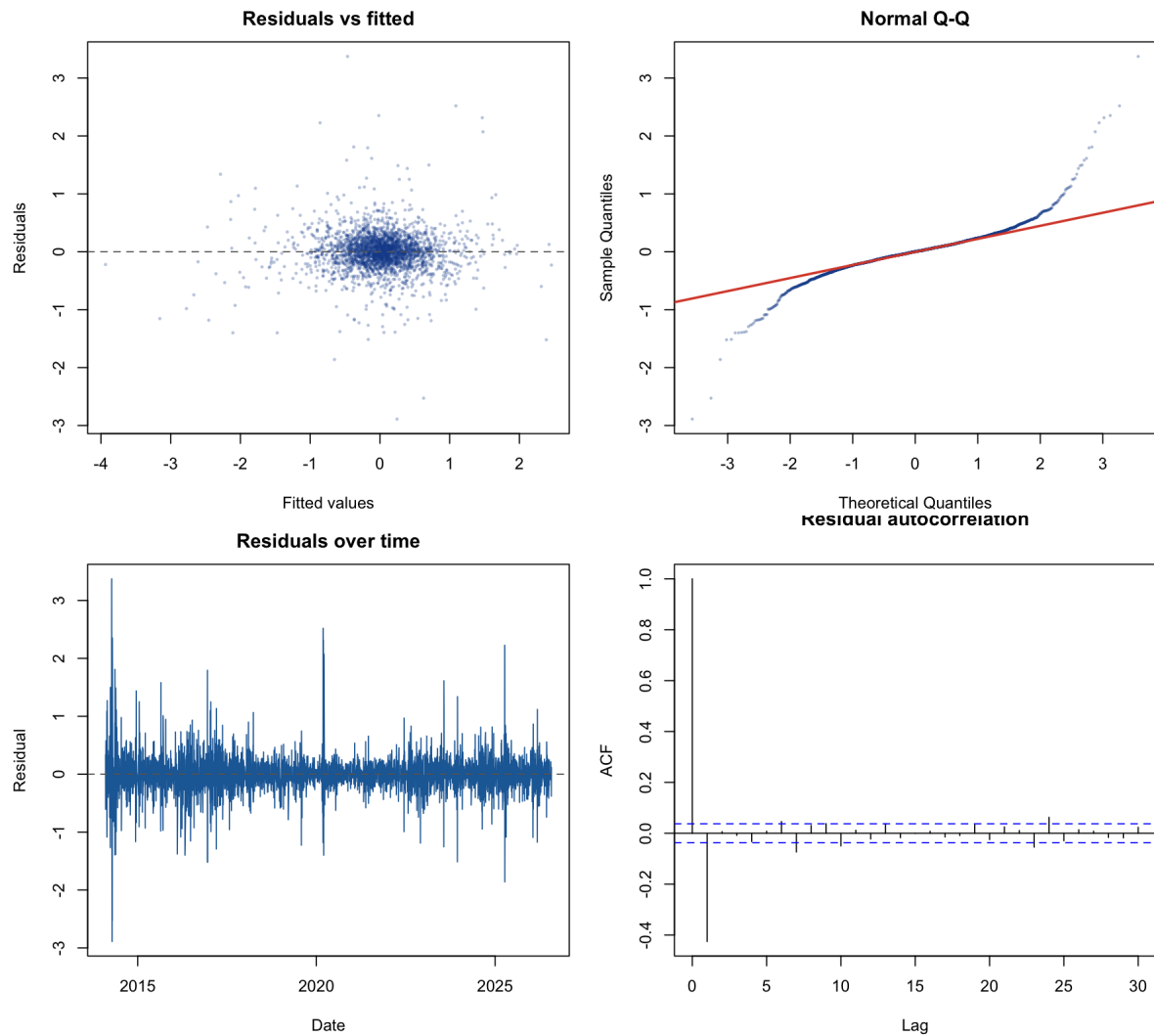


Figure 2. Residual diagnostics motivate robust inference and sensitivity checks.

## Plan for the remaining analysis (275 words)

First, we will make the OLS specification and its estimand explicit: the interaction estimates a conditional change in the contemporaneous yen-spread association, not a causal pandemic effect. Classical OLS intervals will be shown for course continuity, while Newey-West HAC(5) intervals will be the primary time-series sensitivity analysis. We will also report the two implied period-specific yen slopes and formally test each against -1.

Second, we will investigate misspecification identified in the preliminary diagnostics. Residual-versus-fitted, Q-Q, time-order, and autocorrelation plots will be included. We will examine influential observations using Cook's distance and repeat the focal estimates after pre-specified winsorization of continuous daily changes at the 0.5th and 99.5th percentiles. This robustness analysis will be labeled as secondary; it will not replace the unmodified-data result. We will compare HAC lags of 1, 5, and 10 trading days and assess whether conclusions about the interaction change.

Third, we will test whether the selected break date drives the result by using a small, declared sensitivity window around March 11, 2020, rather than searching over all possible dates. We will also compare the full model with the focal-only model using adjusted ( $R^2$ ), residual behavior, and the stability of the focal coefficients. Variance-inflation factors will document collinearity.

Finally, prediction will remain secondary to explanation. We will preserve temporal ordering, compare against a historical-mean benchmark, and report RMSE and MAE on the untouched final 20% of dates. No random cross-validation will be used for time-series observations. All final tables and figures will be generated by the supplied R code from the frozen raw files, and discrepancies between conventional and robust inference will be reported rather than selectively omitted.

## References

Campbell, J. Y., Serfaty-de Medeiros, K., & Viceira, L. M. (2010). Global currency hedging. *The Journal of Finance*, 65(1), 87-121. <https://doi.org/10.1111/j.1540-6261.2009.01524.x>

Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1-22. <https://doi.org/10.1016/j.jfineco.2014.10.010>

Hau, H., & Rey, H. (2006). Exchange rates, equity prices, and capital flows. *The Review of Financial Studies*, 19(1), 273-317. <https://doi.org/10.1093/rfs/hhj008>

Shank, C. A., & Vianna, A. C. (2016). Are US-dollar-hedged-ETF investors aggressive on exchange rates? A panel VAR approach. *Research in International Business and Finance*, 38, 430-438. <https://doi.org/10.1016/j.ribaf.2016.05.002>

## Data and product documentation

Federal Reserve Bank of St. Louis. DEXJPUS; NIKKEI225; VIXCLS; IRSTCI01USM156N; IRSTCI01JPM156N. FRED. iShares. HEWJ and EWJ fund pages and currency-hedged product brief. Kenneth R. French Data Library. Japan 5 Factors [Daily] and Japan Momentum Factor [Daily]. World Health Organization. (2020, March 11). WHO Director-General's opening remarks at the media briefing on COVID-19.

## Submission items still requiring group input

1. Replace bracketed member names and contribution statements.
2. Make the separate contribution agreement PDF match this proposal.
3. Upload the cleaned CSV and raw-source folder to OneDrive and insert shareable links if the course submission form requires them.

## 第二部分 - 中文提案（组内讨论与理解版本）

正式课程提交以第一部分英文版本为准；本部分与英文版本使用相同的研究设计、数值和限制。

### 成员贡献说明

| 成员          | 计划贡献                     |
|-------------|--------------------------|
| Haiwen Yi   | 研究设计、数据构建、R 分析、诊断检验和初稿撰写 |
| [成员 2]      | [补充文献、验证、写作或展示任务]        |
| [成员 3]      | [补充文献、验证、写作或展示任务]        |
| [成员 4, 如适用] | [补充具体贡献]                 |

姓名与贡献说明必须和单独提交的小组协议一致。

### 研究背景与问题

国际股票投资者面对两类回报来源：本地股票市场的涨跌，以及把海外资产换算为本币时产生的汇率变动。货币对冲 ETF 试图剔除第二种影响，但由于远期合约需要定期展期、不同市场交易日不完全一致、费用与执行成本存在，而且投资组合并非连续再平衡，对冲效果不一定完整。本项目研究：日元升值在多大程度上解释货币对冲型日本股票 ETF（HEWJ）与非对冲型 ETF（EWJ）的日度收益差？这种关系在世界卫生组织宣布 COVID-19 构成全球大流行之后是否发生变化？

响应变量定义为日度对数收益差： $(Y_t = 100[\Delta \log(\text{HEWJ}_t) - \Delta \log(\text{EWJ}_t)])$ 。核心预测变量为  $(\text{JPYapp}_t = -100 \Delta \log(\text{DEXJPUS}_t))$ ，因此正值表示日元相对美元升值。如果货币对冲完全消除了同期汇率敞口，而且两个基金除此之外跟踪同一日本股票组合，那么理论斜率应接近 -1：日元升值应使未对冲基金相对于对冲基金获得更高收益。模型使用一个分类时期变量及其与日元升值的交互项，分别估计疫情前后的斜率。2020 年 3 月 11 日作为转换日被剔除，疫情后时期从 3 月 12 日开始。

研究设计基于四篇经过核验的同行评议论文。Hau 与 Rey（2006）讨论汇率、股票回报和投资组合再平衡之间的联系；Campbell、Serfaty-de Medeiros 与 Viceira（2010）说明最优货币对冲会随投资者和资产环境变化；Fama 与 French（2015）为规模、价值、盈利和投资因子控制提供依据；Shank 与 Vianna（2016）研究美国上市货币对冲 ETF 交易与汇率变动之间的动态关系。本项目的贡献更窄但更透明：针对一组日本对冲/非对冲 ETF 进行可复现的日度配对分析，加入明确的结构变化交互项、市场风险控制和时间序列稳健推断。疫情指标仅用于描述结构变化，不被解释为因果处理，因为同期还有许多政策和市场环境同时变化。

### 数据说明

分析样本包含 2,753 个完整日度观测，时间范围为 2014 年 2 月 6 日至 2026 年 7 月 31 日，其中疫情前 1,348 个、疫情后 1,405 个。HEWJ 和 EWJ 的复权收盘价来自 Yahoo Finance 公开历史行情接口。两只基金构成较自然的配对：iShares 将 HEWJ 描述为在获得日本股票敞口的同时对冲日元风险，并把 EWJ 列为对应的未对冲日本股票基金。DEXJPUS（日元/美元汇率，纽约中午报价）来自圣路易斯联储 FRED。

响应变量为 HEWJ 与 EWJ 的日度对数收益差。三个核心模型项为日元升值、分类变量 Post 以及  $\text{JPYapp} \times \text{Post}$ 。另外八个预测变量用于控制可观测市场环境：日经 225 日收益、日本 SMB、HML、RMW 和 CMA 因子、日本动量因子 MOM、VIX 日度对数变化，以及滞后一个月的美国与日本短期利率差。日本因子来自 Kenneth French 公开数据库并以美元计量；日经 225、VIX 和利率序列来自 FRED。利率差滞后一个月，避免把月初尚无法获知的当月数值用于日度解释。

所有序列先按交易日对齐，再计算变化率，最后保留完整观测。百分比变量以百分点表示，使核心斜率可以直接解释。仓库保存了冻结原始文件、SHA-256 校验值、清理后的 CSV、R 脚本以及机器可读的模型结果。数据入口包括 Yahoo Finance 的 HEWJ/EWJ 历史页、FRED 的 DEXJPUS/NIKKEI225/VIXCLS/利率序列，以及 Kenneth French Data Library。

## 伦理声明

本项目只使用公开、聚合的市场价格和因子序列，不包含个人信息、人体研究数据、私人持仓或个人交易记录，因此隐私和知情同意风险很低。不过，如果把描述性关联写成投资建议或疫情的因果效应，金融研究结果仍可能被误用。报告会明确说明：结论只描述该 ETF 配对在本样本期间的表现，不能证明 COVID-19 导致斜率变化。还存在存续偏差和产品选择问题：选择 HEWJ/EWJ 是因为它们构成方便的配对，而不是因为它们代表所有货币对冲产品的随机样本。

可复现性通过保存精确下载文件、记录变量变换、明确剔除 2020 年 3 月 11 日并在 R 中执行完整分析来保证。复权价格包含数据提供方定义的公司行动调整，不应被解释为不含交易成本即可成交的价格。Yahoo、FRED、iShares 和 Kenneth French 数据仍受各自发布方条款和修订机制约束；项目会引用来源和分享入口，不宣称拥有这些数据。历史预测评估也不构成未来收益保证或交易建议。

## 初步结果

| 指标          | 结果      | 指标         | 结果              |
|-------------|---------|------------|-----------------|
| 完整模型 $R^2$  | 71.21%  | 疫情前日元斜率    | -0.852          |
| 日元 x 疫情后交互项 | -0.0648 | 经典 / HAC p | 0.0043 / 0.0683 |
| 疫情后隐含斜率     | -0.917  | 最大 VIF     | 3.55 (HML)      |

完整普通最小二乘模型为：

$$Y_t = \beta_0 + \beta_1 JPY_{app,t} + \beta_2 Post_t + \beta_3 (JPY_{app,t} \times Post_t) + \gamma' Z_t + \varepsilon_t,$$

其中 ( $Z_t$ ) 包含八个市场与因子控制变量。完整模型解释了 HEWJ-EWJ 日度收益差 71.21% 的变异（调整后  $R^2=71.10\%$ ）；仅含核心变量的模型解释 70.37%。疫情前日元升值斜率为 -0.852（经典标准误 0.0178；HAC(5) 标准误 0.0307；HAC  $p < 0.001$ ）。也就是说，在控制其他变量后，疫情前日元每升值 1 个百分点，HEWJ 相对 EWJ 的日度收益平均降低约 0.852 个百分点。

交互项估计为 -0.0648。使用经典 OLS 标准误时显著（标准误 0.0227， $p = 0.0043$ ），但使用 Newey-West HAC(5) 后仅为边际显著（标准误 0.0355， $p = 0.0683$ ，95% 区间 [-0.1345, 0.0049]）。由此得到疫情后隐含斜率 -0.917。HAC(5) 检验表明，疫情前后两个斜率都与理论基准 -1 存在统计差异，但疫情后的经济差距更小。在日元变化为零时，时期水平差为 0.0085 个百分点，HAC  $p = 0.176$ ，并不显著。控制变量中，VIX 变化系数为负（-0.00687，HAC  $p < 0.001$ ），其他多数控制项估计不精确。

模型并不满足理想的独立同分布误差假设。Breusch-Pagan 检验拒绝同方差（ $p = 0.0048$ ）；五阶 Breusch-Godfrey 检验拒绝无序列相关（ $p < 0.001$ ）；Q-Q 图显示厚尾。Durbin-Watson 统计量为 2.85，与残差的一阶负自相关一致。多重共线性不严重，最大 VIF 为 3.55（HML）。因此，报告同时给出经典 OLS 与 HAC(5) 不确定性，而不是只依赖常规  $p$  值。

作为次要预测检验，样本按时间顺序做 80/20 划分：训练集截至 2024 年 1 月 29 日，测试集为 2024 年 1 月 30 日至 2026 年 7 月 31 日。完整模型的 RMSE 为 0.313、MAE 为 0.219 个百分点；训练期历史均值基准分别为 0.654 和 0.463。该结果提供辅助支持，但不能替代统计推断，因为较晚观测可能来自不同市场状态。

## 后续分析计划

### 第一，明确 OLS

规格和估计对象：交互项估计的是日元与收益差同期关联的条件变化，不是疫情的因果效应。课程主体继续展示经典 OLS 区间，同时把 Newey-West HAC(5) 作为主要时间序列稳健性分析，并报告疫情前后两个隐含斜率及其相对于 -1 的正式检验。

第二，根据初步诊断调查模型错误。报告将包含残差-拟合值图、Q-Q 图、按时间排列的残差图和自相关图；用 Cook 距离识别有影响力观测；并在连续日变化变量的 0.5% 和 99.5%

分位数进行预先声明的缩尾敏感性分析。该分析会明确标注为次要结果，不替代未经修改的数据结果。同时比较 1、5 和 10 个交易日的 HAC 滞后长度，检查交互项结论是否变化。

第三，在 2020 年 3 月 11 日附近使用一个事先声明的小范围日期敏感性窗口，判断结果是否完全由所选断点驱动，而不是在所有候选日期中事后搜寻最显著断点。完整模型与核心变量模型将通过调整后 ( $R^2$ )、残差行为和核心系数稳定性进行比较，并继续报告 VIF。

最后，预测仍然从属于解释目标。验证必须保持时间顺序，与历史均值基准比较，并在未触碰的最后 20% 日期上报告 RMSE 和 MAE。时间序列观测不使用随机交叉验证。所有最终表格和图形都由仓库中的 R 代码从冻结原始文件生成；经典推断与稳健推断之间的差异会如实保留。

## 参考文献

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## 提交前仍需补充

1. 替换所有带方括号的小组成员姓名和贡献说明。
2. 确保单独提交的贡献协议 PDF 与本提案一致。
3. 把清理后的 CSV 与原始文件上传至 OneDrive，并按课程表单要求填入可访问链接。